

FEA-MER: A Framework Based on Fusion-First and Emotion-Guided Alignment for Multimodal Emotion Recognition

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Abstract

Most existing multimodal emotion recognition (MER) approaches follow an “alignment-first, fusion-later” paradigm, which first aligns cross-modal features and then models their joint distribution. However, as different emotions depend on different modalities to varying extents, this approach fails to capture the emotional relationships between modalities, which undermines recognition performance. Consequently, early alignment may suppress modality-specific emotional cues, impairing classification accuracy. To address this, we propose **FEA-MER** (Fusion-First and Emotion-Guided Alignment for Multimodal Emotion Recognition), a novel framework that adopts a “fusion-first, alignment-later” strategy to better preserve and integrate emotion-relevant information from different modalities. FEA-MER is structured into four main modules: (1) Modality-Specific Encoding for extracting contextual features from speech and text; (2) Fusion Encoding, which performs intra- and inter-modal interaction via a Transformer-based self-attention mechanism; (3) Emotion-Aware Optimization, incorporating speech-text matching and contrastive learning to reinforce modality alignment and emotional disentanglement; and (4) Emotion Classification, which maps the final representation to discrete categories. These components work jointly to enhance emotion-aware representation learning and improve classification performance. Experiments on IEMOCAP and MELD demonstrate that FEA-MER achieves competitive or state-of-the-art results, confirming the effectiveness of our proposed fusion-first and alignment-guided paradigm. Our source code and pre-trained models are available at: <https://anonymous.4open.science/r/FEA-MER-8EDE>.

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Learning latent representations*.

Keywords

Multimodal Emotion Recognition, Representation Learning, Continuous Emotions, Contrastive Learning

1 Introduction

The Multimodal Emotion Recognition (MER) based on speech and text aims to perceive the affective attitude of humans by integrating acoustic and semantic cues [1]. It has found increasing importance in a variety of emerging applications, such as online chatting, news analysis, and dialogue systems [2, 3]. Predicting emotions accurately from multimodal data remains challenging, largely due to the complexity and inherent ambiguity of human emotional expressions [4, 5]. Unlike unimodal emotion recognition approaches, multimodal emotion recognition leverages complementary emotional cues across modalities, thus achieving a richer and more nuanced understanding of emotional states [6]. However, effective

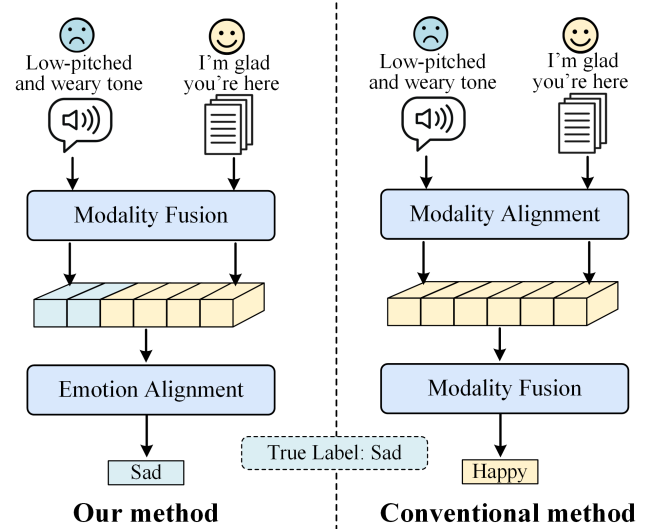


Figure 1: Illustration of Alignment-First vs. Fusion-First Strategies in MER. Our approach preserves modality-specific discrepancies before emotion-aware alignment, mitigating misalignment in emotionally inconsistent cases.

utilization of multimodal cues requires learning the dominant emotional signals from each modality, modeling their interactions in a coordinated manner, and ultimately mapping the fused representation to the correct emotion category [7].

A milestone in the field of multimodal representation learning is the introduction of ALBEF (Align Before Fuse) [8, 9], which demonstrated that aligning visual and textual features prior to fusion can significantly enhance downstream performance. Inspired by this idea, a growing body of work in MER has adopted a similar “alignment-first, fusion-later” paradigm, which first reduces the modality gap and then leverages fused features for emotion prediction [10–12]. In this paradigm, alignment mitigates cross-modal heterogeneity to improve fusion reliability, while fusion integrates complementary modal features to facilitate downstream fine-grained emotion classification tasks.

In the task of MER, existing alignment techniques can be broadly classified into two categories: contrastive learning-based alignment and subspace-based alignment [1, 2]. Among them, contrastive learning-based methods aim to enhance semantic consistency between modalities by drawing positive sample pairs closer while pushing negative pairs further apart [13, 14]. However, in the context of MER, a key challenge arises from the fact that emotional cues are unevenly distributed across modalities—different emotions often depend on different modalities for accurate recognition.

Consequently, enforcing uniform semantic alignment may inadvertently obscure modality-specific affective signals that are critical for emotion classification, ultimately impairing the model's ability to capture nuanced emotional expressions and reducing overall performance. To address the limitations of uniform alignment, some methods adopt dual-space alignment, which separates features into shared and private subspaces [15, 16]. This design enables the model to capture cross-modal emotional semantics in a shared space while preserving modality-specific cues in private spaces. Despite their effectiveness in preserving modality-specific cues, these methods are difficult to optimize and may introduce irrelevant noise due to improper separation of task-relevant shared and private features.

On the fusion side, a wide range of techniques have been proposed to effectively integrate emotional cues from different modalities. Among them, attention-based fusion mechanisms—especially cross-attention—have attracted significant attention due to their ability to capture fine-grained inter-modal interactions [17, 18]. However, such methods often rely on manually designed query strategies tailored to specific modality characteristics, making them sensitive to modality imbalance and less adaptable across diverse datasets.

To address the aforementioned challenges, we propose a novel multimodal emotion recognition framework, termed Fusion-First and Emotion-Guided Alignment for Multimodal Emotion Recognition (FEA-MER). In contrast to conventional alignment-first paradigms, as illustrated in Figure 1, our framework follows a "fusion-first, alignment-later" strategy, aiming to retain modality-specific emotional cues during the early stages of processing while enhancing semantic consistency in a later, emotion-guided manner.

First, we design a self-attention-based encoding module that integrates multi-modal information without relying on hand-crafted cross-modal query strategies. Additionally, inspired by the success of auxiliary objectives in vision-language tasks—such as image-text retrieval, visual entailment, and visual question answering (VQA) [8, 19]—we introduce an auxiliary multimodal emotion matching task to enhance the fusion module's ability to capture complementary emotional cues. This task encourages the fusion module to model the synergy between dominant emotions across modalities, thereby enhancing the emotion-discriminative capacity of the fused representation and indirectly improving the downstream classification performance.

Second, to reduce the classification difficulty in the affective space, we propose an emotion-guided weighted contrastive learning strategy that explicitly aligns fused representations based on emotional similarity. Unlike prior contrastive learning methods that treat all sample pairs equally, we incorporate emotional intensity as a continuous soft weight to dynamically adjust sample contributions during training. This allows the model to focus on harder samples, yielding more fine-grained and structured emotional representations, and enhancing downstream classification performance.

In summary, the main contributions of our work are as follows:

- We introduce a novel fusion-first modeling paradigm, which better preserves modality-specific emotional cues and alleviates early alignment issues.

- We introduce a representation optimization mechanism that explicitly models emotion-level consistency and semantic structure in the fused space, enhancing the discriminability of multimodal affective representations.
- Experiments on two benchmark datasets, IEMOCAP and MELD, show that our approach delivers competitive performance, demonstrating its effectiveness and generalization capability.

2 Related Work

2.1 Modality Alignment

Modality alignment is a critical step in MER, aiming to reduce the semantic and representational discrepancies between modalities such as speech and text, thereby facilitating effective information fusion. Current alignment techniques can broadly be classified into two primary categories: contrastive learning-based alignment and subspace-based alignment.

Contrastive learning has emerged as a powerful paradigm for aligning heterogeneous modality representations in multimodal tasks, including MER. Broadly, contrastive learning methods can be categorized into two types: unsupervised contrastive learning and supervised contrastive learning, which differ mainly in the availability and usage of label information. Unsupervised contrastive learning methods—such as SimCLR [20], MoCo [21], and InfoNCE-based frameworks [22]—construct positive pairs from different augmentations of the same instance and treat other samples as negatives, aiming to learn generalizable and instance-discriminative representations. These approaches are predominantly employed in vision and language pretraining, where task labels are unavailable. By contrast, supervised contrastive learning (SCL) [23] incorporates label information to explicitly pull together samples from the same class and push apart those from different classes. SCL has been increasingly adopted in MER due to the availability of emotional labels and the necessity of class-level discrimination. For example, Zheng et al. [24] designed a weakly supervised contrastive framework to handle class collision problems. Similarly, recent MER models such as Ghosh et al. [25] and Sun et al. [26] have integrated SCL-based objectives (e.g., modality matching, masked prediction) to enhance inter-modal semantic consistency and robustness. Despite these advancements, the theoretical motivation of contrastive learning fundamentally stems from the InfoMax principle—i.e., maximizing the mutual information (MI) between two correlated views [27]. However, in MER, different modalities contribute unequally to different emotional categories, and emotional cues are often modality-specific (e.g., sad in speech vs. happy in text). Therefore, enforcing strong alignment through MI maximization at early fusion stages can oversuppress discriminative, modality-specific affective signals, thus degrading the emotion classification performance.

Unlike contrastive learning methods that focus on unifying multimodal representations, subspace alignment approaches divide features into shared subspaces (modality-invariant) and private subspaces (modality-specific) to simultaneously model inter-modal consistency and intra-modal divergence, thereby avoiding the suppression of discriminative emotional cues caused by over-alignment. Representative methods include MISA [11], MFM [28], and hierarchical alignment models [26], which enhance model robustness

and interpretability through feature disentanglement. Although such methods are effective in preserving modality-specific information, they often rely on complex architectures (e.g., multi-stream encoders, orthogonality constraints, or adversarial objectives), making it difficult to balance shared and private representations during training. This can lead to optimization instability or redundant noise, thereby limiting their practicality and scalability.

2.2 Modality Fusion

Modality fusion aims to integrate emotional cues from different modalities to construct more discriminative joint representations, thereby improving the performance of MER. Based on the stage at which fusion occurs, existing fusion methods can be broadly categorized into three types:

Early Fusion: This approach concatenates raw features from multiple modalities at the input level. Although structurally simple and computationally efficient, it is highly sensitive to scale differences and noise across modalities, often resulting in redundant or conflicting information [29]. **Late Fusion:** Each modality is processed independently for emotion classification, and the final predictions are combined at the decision level, typically through weighted averaging or voting. While robust to noise, this method fails to capture deep semantic interactions across modalities [30, 31]. **Intermediate Fusion:** Currently the dominant paradigm, this strategy encodes each modality separately before performing semantic-level fusion. It strikes a balance between representational power and interaction modeling [32].

From the perspective of fusion mechanisms, several mainstream techniques have been explored: **Concatenation:** A simple and commonly used method, but it lacks the capacity to capture inter-modality dependencies. **Tensor-based Fusion:** Methods such as TMOPFN [33] and PFBT-STP [34] model higher-order interactions, but suffer from high computational costs. **Attention-based Fusion:** This has become the dominant strategy in recent years due to its ability to dynamically select and weigh informative features. Among them, the cross-attention mechanism has been widely applied in MER tasks to enable fine-grained alignment and interaction between modalities. For example, Khan et al. [35] proposed a novel model that utilizes a multi-headed cross-attention mechanism for deep feature fusion, enhancing the interaction between audio and text cues and significantly improving emotion recognition performance. However, cross-attention approaches suffer from two key limitations: (1) reliance on manually crafted, dataset-dependent query strategies, which undermines generalization; and (2) susceptibility to modality dominance, where stronger modalities may overshadow critical cues from weaker ones [36, 37].

To address the above challenges, some studies have explored the application of self-attention mechanisms in multimodal fusion to achieve more symmetric and unified cross-modal interaction modeling. Compared to cross-attention methods that rely on manually designed query strategies, self-attention-based fusion frameworks built upon Transformers offer greater flexibility while achieving comparable performance in downstream tasks [38], [39].

Furthermore, inspired by the success of multi-task learning in vision-language tasks (e.g., VQA, image-text retrieval, and visual entailment), several MER studies have incorporated auxiliary

tasks—such as modality matching, masked prediction, and contrastive learning—into the fusion process. These tasks explicitly guide cross-modal alignment and interaction, thereby enhancing the quality of the fused representations and improving downstream performance [40, 41].

3 Methodology

3.1 Problem Definition

We define notations for FEA-MER. Let $X^a \in \mathbb{R}^{n_a \times d_a}$ and $X^t \in \mathbb{R}^{n_t \times d_t}$ denote the heterogeneous feature spaces for acoustic (a) and textual (t) modalities, respectively, where n_m and d_m represent the sequence length and feature dimension for modality $m \in \{a, t\}$. Let $\mathcal{Y} = \{1, 2, \dots, C\}$ denote the label space with C emotion categories.

Given a training dataset $\mathcal{D} = \{(X_i^a, X_i^t, y_i)\}_{i=1}^N$, where (X_i^a, X_i^t) are the acoustic and textual inputs of the i -th utterance, and $y_i \in \mathcal{Y}$ is the corresponding emotion label, FEA-MER aims to learn a mapping function:

$$\mathcal{F} : X^a \times X^t \rightarrow \mathcal{Y}$$

that predicts the emotion category associated with each utterance by integrating complementary cues from different modalities.

In particular, FEA-MER independently encodes each modality, fuses them via a self-attention module, and introduces an auxiliary STM task to enhance cross-modal collaboration. An emotion-guided alignment module then refines the fused representation based on emotional similarity for final classification. The predicted label is given by:

$$\hat{y}_i = \arg \max_j p(y_i = j \mid X_i^a, X_i^t)$$

3.2 Model Overview

As illustrated in Figure 2, **FEA-MER** consists of four main modules: (1) *Modality-Specific Encoding*, (2) *Fusion Encoding*, (3) *Emotion-Aware Optimization*, and (4) *Emotion Classification*.

(1) Modality-Specific Encoding. We utilize pre-trained encoders to extract semantic features from each modality. For the speech stream, we adopt Wav2Vec2.0-Large fine-tuned on the MSP-Podcast dataset [42]. For textual input, we use RoBERTa-Large. Given an utterance pair (X^a, X^t) , each encoder transforms the input into contextual embeddings, which are concatenated into a unified sequence with positional encoding and modality-aware attention masks.

(2) Fusion Encoding. The concatenated sequence is fed into a Transformer-based fusion module composed of M stacked encoder blocks, each containing multi-head self-attention and feed-forward layers. This design enables symmetric, scalable modeling of intra- and inter-modal interactions without relying on modality-specific query construction.

(3) Emotion-Aware Optimization. We introduce two auxiliary tasks to enhance emotional discrimination:

- **Speech-Text Matching (STM):** This is a binary classification task that determines whether a given speech-text pair is emotionally consistent. It aims to enhance emotional coordination across modalities, thereby improving the fusion module’s modeling capacity and indirectly boosting the performance of the primary emotion classification task.

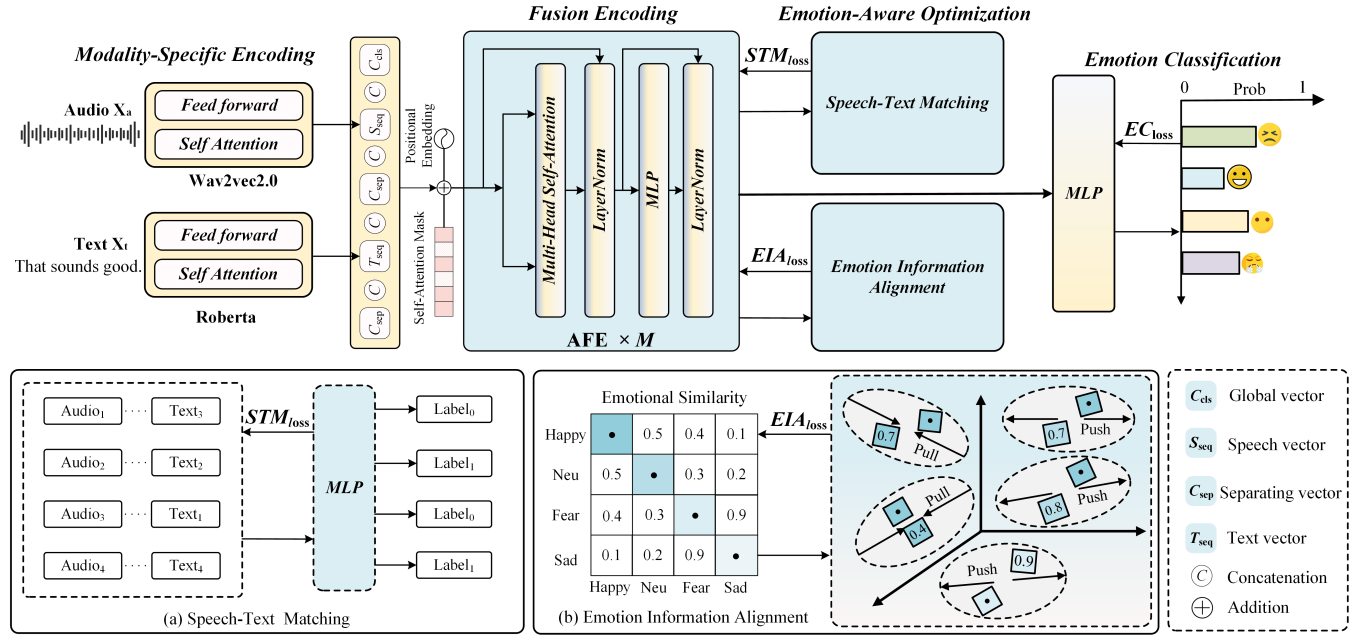


Figure 2: Architecture of the Proposed FEA-MER Framework. It sequentially comprises four modules: modality-specific encoding, self-attention-based fusion, emotion-aware optimization (STM & EIA), and emotion classification.

- **Emotion Information Alignment (EIA):** A contrastive objective that leverages continuous emotion embeddings to guide fine-grained alignment of representations based on emotional similarity.

(4) **Emotion Classification.** Finally, a multi-layer perceptron (MLP) maps the fused representation to discrete emotion categories for classification.

3.3 Self-attention-based fusion module

To enable our proposed self-attention-based multimodal fusion module to perform adaptive query fusion across different datasets, we concatenate the multimodal input sequences as shown in Equation 1.

$$X_{fusion} = f_{FE}(\text{Concat}(C_{cls}, S_{seq}, C_{sep}, T_{seq}, C_{sep})) \quad (1)$$

Where, $\text{Concat}(\cdot)$ denotes the concatenation function, C_{cls} and C_{sep} represent the classification token vector and separator token vector, respectively. C_{cls} is designed to aggregate global information from the input features during the modality fusion process, while C_{sep} acts as a boundary to distinguish between the two modalities. Finally, $f_{FE}(\cdot)$ processes the concatenated sequence to generate the fused feature vector X_{fusion} .

3.4 Auxiliary branches for emotion-aware learning

3.4.1 **Speech-Text Matching.** In addition, to promote cross-modal collaborative modeling, we introduce an STM task. By constructing pseudo-paired samples with corresponding labels, this task guides the fusion encoder to retain emotion-discriminative information

during cross-modal integration. It helps prevent dominant modalities from overwhelming complementary emotional cues from other modalities, thereby ensuring accurate multimodal emotion recognition even in cases of emotional inconsistency across modalities.

Specifically, for each training batch, we randomly recombine speech and text samples from different utterances to form new multimodal pairs. If the original emotion labels of both modalities are the same, the pair is labeled as emotionally matched ($y^m = 1$); otherwise, it is labeled as mismatched ($y^m = 0$).

Finally, we use a linear classification head for emotion information decoupling. The loss function $\mathcal{L}_{STM}$ is optimized using cross-entropy, as defined in Equation 2:

$$\mathcal{L}_{STM} = -[y^m \log(\hat{y}^m) + (1 - y^m) \log(1 - \hat{y}^m)] \quad (2)$$

Here, y^m is the ground-truth binary label indicating whether the emotional semantics of the speech-text pair match (1 for matching, 0 otherwise), and $\hat{y}^m$ is the predicted probability computed via a sigmoid function.

3.4.2 **Emotion Information Alignment.** To enhance the emotional structure of the multimodal representation space, we propose an Emotion-Guided Weighted Contrastive Learning (EIA) strategy. Unlike traditional methods that treat all positive and negative pairs equally, our method introduces continuous emotion vectors to compute pairwise similarity and apply soft weights accordingly, encouraging the model to construct an embedding space that captures both category distinctions and emotion-level similarity.

Specifically, we fine-tune a pre-trained speech encoder to extract 3-dimensional continuous emotion vectors e_i (e.g., valence, arousal, dominance) for each sample i in a batch of size N . Contrastive

learning is performed over the fused multimodal representation $\mathbf{h}_i$, after speech-text fusion.

For each anchor $\mathbf{h}_i$, let $\mathcal{P}_i$ and $\mathcal{N}_i$ denote the sets of indices corresponding to its positive and negative samples, respectively.

We define the pairwise emotion similarity as:

$$\text{sim}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \cdot \|\mathbf{e}_j\|} \quad (3)$$

We then assign weights based on emotion similarity:

- For positive pairs ($j \in \mathcal{P}_i$), a higher weight is assigned to emotionally closer pairs:

$$w_{ij}^+ = \frac{1}{\epsilon + 1 - \text{sim}(\mathbf{e}_i, \mathbf{e}_j)} \quad (4)$$

- For negative pairs ($j \in \mathcal{N}_i$), a higher weight is assigned to emotionally distant pairs:

$$w_{ij}^- = \epsilon + 1 - \text{sim}(\mathbf{e}_i, \mathbf{e}_j) \quad (5)$$

Here, $\epsilon = 1 \times 10^{-8}$ is a small constant for numerical stability.

To project the fused features $\mathbf{h}_i$ into a contrastive embedding space, we apply a two-layer projection head $f(\cdot)$:

$$\mathbf{z}_i = f(\mathbf{h}_i) = W_2 \cdot \text{ReLU}(W_1 \mathbf{h}_i + \mathbf{b}_1) + \mathbf{b}_2 \quad (6)$$

Here, $W_1 \in \mathbb{R}^{d \times d_h}$ and $W_2 \in \mathbb{R}^{256 \times d}$ denote the learnable projection matrices, and $\mathbf{b}_1 \in \mathbb{R}^d$, $\mathbf{b}_2 \in \mathbb{R}^{256}$ are the corresponding bias terms. The output $\mathbf{z}_i \in \mathbb{R}^{256}$ represents the contrastive embedding, which is then ℓ_2 -normalized before computing cosine similarity.

The weighted contrastive loss for anchor i is computed by first aggregating the weighted similarities:

$$S_i^+ = \sum_{j \in \mathcal{P}_i} w_{ij}^+ \cdot \exp\left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}_j)}{\tau}\right) \quad (7)$$

$$S_i^- = \sum_{j \in \mathcal{N}_i} w_{ij}^- \cdot \exp\left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}_j)}{\tau}\right) \quad (8)$$

Then, the contrastive loss for each anchor sample i is defined as:

$$\mathcal{L}_i = -\log\left(\frac{S_i^+}{S_i^+ + S_i^-}\right) \quad (9)$$

Finally, the overall contrastive loss $\mathcal{L}_{\text{EIA}}$ is computed by averaging over all N samples in the batch:

$$\mathcal{L}_{\text{EIA}} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_i \quad (10)$$

3.5 Emotion classification

After multimodal fusion and auxiliary optimization, the final fused representation $\mathbf{X}_{\text{fusion}}$ is used for discrete emotion classification.

(1) Emotion Classification Head. This module performs C -way classification over the fused representation $\mathbf{X}_{\text{fusion}}$ to predict discrete emotion categories. Let $\mathbf{y} \in \mathbb{R}^C$ be the one-hot ground-truth label vector, and let $\hat{\mathbf{y}} \in \mathbb{R}^C$ denote the predicted class probabilities. Each element $\hat{y}^i$ of the predicted vector is computed as:

$$\hat{y}^i = \frac{\exp\left((W_f \cdot \mathbf{X}_{\text{fusion}} + \mathbf{b}_f)^i\right)}{\sum_{j=1}^C \exp\left((W_f \cdot \mathbf{X}_{\text{fusion}} + \mathbf{b}_f)^j\right)} \quad (11)$$

Here, $W_f \in \mathbb{R}^{C \times d_h}$ and $\mathbf{b}_f \in \mathbb{R}^C$ are the parameters of the classification head, and $\hat{y}^i$ denotes the predicted probability of the i -th emotion class.

The classification loss is computed using standard cross-entropy:

$$\mathcal{L}_{\text{EC}} = - \sum_{i=1}^C y^i \log(\hat{y}^i) \quad (12)$$

(2) Overall Objective. The full training objective combines three components: (i) the emotion-guided contrastive loss $\mathcal{L}_{\text{EIA}}$, (ii) the emotion classification loss $\mathcal{L}_{\text{EC}}$, and (iii) the auxiliary speech-text matching loss $\mathcal{L}_{\text{STM}}$. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{EIA}} + \beta \cdot \mathcal{L}_{\text{EC}} + \gamma \cdot \mathcal{L}_{\text{STM}} \quad (13)$$

Where α , β , and γ are hyperparameters that balance the contributions of the three objectives.

4 Experiment Settings

This section presents the experimental settings adopted to evaluate the effectiveness of our proposed FEA-MER framework. We first introduce the benchmark datasets employed in our experiments, followed by implementation details including model configurations and training strategies. Next, we describe the baseline models used for performance comparison. Finally, we report quantitative results along with ablation studies to validate the contribution of each component in FEA-MER.

4.1 Datasets

We validated our proposed model on two public multimodal datasets, IEMOCAP [43] and MELD [44], and the label distribution of these datasets is shown in Figure 5.

Specifically, IEMOCAP is a recorded dialogue dataset comprising 12 hours of data, including video, speech, transcriptions, and motion capture of the actors. The dialogues consist of five pairs of speakers, totaling five males and five females. Emotion labels include anger, happiness, sadness, frustration, excitement, fear, surprise, disgust, and others, with three annotators providing ratings for emotional valence, activation, and dominance. For comparison with previous work, we extracted four emotional categories: happiness, sadness, anger, and neutrality, with excitement merged into the happiness category, resulting in a total of 5,531 samples. The experiments utilized a five-fold leave-one-session-out strategy (with eight speakers used for training and two speakers in one session for testing). Unweighted accuracy (ACC, %) and weighted accuracy (WA, %) were employed to measure model performance.

MELD is a challenging multi-party dialogue dataset containing approximately 13,000 utterances from the TV show "Friends," annotated with seven emotion labels (anger, disgust, fear, happiness, neutrality, sadness, and surprise). The modalities provided include speech, visual, and text data. Unlike IEMOCAP, this dataset is divided into training, development, and test sets, allowing for a consistent training and testing strategy for the model. Weighted accuracy (WACC, %) and weighted F1 score (WF1, %) were used to evaluate the model's results.

4.2 Implementation Details

Our model is implemented using the PyTorch framework, with the AdamW optimizer and a learning rate of $1e-5$. The batch size is set to 4, and the output dimension of the projection head function in the contrastive learning component is 256. The multi-modal fusion layer employs 8 attention heads. The hyperparameters for the loss function, α , β , and γ , are set to 0.2, 0.5, and 0.3, respectively. Our training is conducted on a Linux system using an A100 GPU.

4.3 Baselines

We compare FEA-MER against the following works:

- **CAMSER [45]**: Utilizes multi-head cross-attention and deep fusion to integrate speech and text features.
- **MFGCN [46]**: Proposes a multimodal fusion framework that jointly models intra-modal dependencies and inter-modal complementarity to enhance emotion recognition.
- **SAMS [47]**: Performs local and global semantic alignment across modalities using high-level emotion representations.
- **DCMTM [48]**: Proposes a dual cross-modal Transformer framework that integrates spectral features and self-supervised speech representations with text to enhance multimodal emotion recognition.
- **BERT-TL [49]**: Applies transfer learning to speech and text modalities and integrates ResNet and BERT via late fusion to improve emotion recognition on IEMOCAP.
- **MER-HAN [50]**: Introduces a hybrid attention model that captures emotional cues within and between audio and text modalities.
- **LFLB [51]**: Proposes a stacking ensemble of BERT-based text and CNN-BiLSTM-based speech models to improve emotion recognition accuracy on IEMOCAP.
- **DBT [52]**: Proposes a dual-branch transformer using wav2vec2.0, RoBERTa, and weighted DS fusion with label smoothing for improved multimodal emotion recognition.
- **MCFN [53]**: Proposes a fusion network that enhances emotion recognition by integrating unimodal and cross-modal cues through collaborative learning and dual-stage fusion.
- **MCSCAN [54]**: Employs parallel cross- and self-attention mechanisms for multimodal fusion.
- **DIMMN [55]**: Utilizes dynamic memory integration to model long-range dependencies for emotion recognition.
- **SACMA [56]**: Introduces speaker-aware context modeling to enhance multimodal emotion perception.
- **SMCN [57]**: Proposes a self-guided modality alignment mechanism to improve representation learning.
- **RMERCT [58]**: Applies Transformer-based cross-modal fusion for robust multimodal emotion recognition.
- **SMIN [59]**: Combines semi-supervised learning with modality interaction for emotion classification.
- **CARAT [60]**: Employs contrastive feature reconstruction and aggregation to enhance multimodal emotion recognition.

4.4 Comparative Experiments

To evaluate the effectiveness of the proposed FEA-MER framework, we conduct comprehensive comparisons against a range of

representative baselines on two widely used benchmark datasets: **IEMOCAP** and **MELD**. The results are shown in Tables 1 and 2.

On the **IEMOCAP** dataset, our method achieves state-of-the-art results with an ACC of 80.2% and a WACC of 78.9%, outperforming the strongest baseline **DCMTM** by 1.0% in ACC and 0.4% in WACC. Compared to earlier alignment-based models such as **DBT**, **FEA-MER** achieves a substantial gain of +6.2% in ACC and +6.7% in WACC. Our model also demonstrates superior performance across most individual emotion categories, particularly on *sadness* and *anger*, highlighting its robustness in capturing modality-specific emotional cues.

On the **MELD** dataset, FEA-MER attains the best overall results with a WACC of 66.5% and a WF1 of 65.1%. It exceeds the best baseline **CARAT** by 0.8% and 0.5% in WACC and WF1, respectively. Despite the increased difficulty due to label imbalance and emotional ambiguity in MELD, FEA-MER consistently delivers competitive performance, confirming its strong generalization ability across domains.

In summary, the experimental results across both benchmarks demonstrate that FEA-MER not only enhances global classification performance but also offers improved emotion-specific discrimination, validating the benefits of its fusion-first and emotion-guided alignment design.

4.5 Ablation Studies of FEA-MER

To comprehensively assess the effectiveness of the proposed fusion-first and alignment-later strategy, as well as the contribution of each optimization component, we conduct detailed ablation studies on the IEMOCAP and MELD datasets. The experimental configurations and results are reported in Table 3.

We compare the following model variants:

- **Text-only / Speech-only**: Unimodal baselines using only text or speech features with an MLP classifier.
- **Speech+Text + MLP**: Directly concatenates speech and text embeddings, followed by an MLP without fusion modeling.
- **Cross-Attention Fusion**: Replaces self-attention with bidirectional cross-attention in each fusion encoder block.
- **w/o STM**: Removes the Speech-Text Matching objective ($\mathcal{L}_{STM}$).
- **w/o EIA**: Removes the Emotion Information Alignment objective ($\mathcal{L}_{EIA}$).
- **w/o STM & EIA**: Removes both auxiliary objectives.
- **Align $\rightarrow$ Fuse**: Applies supervised contrastive learning to align modalities before fusion.
- **w/o EIA Weighting**: Removes the weighting in the EIA objective, making the weights of all positive and negative sample pairs equal.
- **FEA-MER**: Our full model with all proposed modules.

From Table 3, we observe several key findings. First, unimodal variants (Text-only or Speech-only) perform significantly worse across all metrics compared to multimodal configurations, validating the benefit of cross-modal integration. Second, simply concatenating multimodal embeddings without interaction modeling yields

Table 1: Model Performance Comparison on the IEMOCAP

Methods	Align → Fuse Order	Hap	Sad	Ang	Neu	ACC	WACC
CAMSER [45]	Align → Fuse	-	-	-	-	77.2	76.5
MFGCN [46]	Align → Fuse	78.9	<u>82.3</u>	83.3	68.2	77.3	78.3
SAMS [†] [47]	Align → Fuse	77.1	81.2	<u>84.7</u>	66.1	77.2	76.3
DCMTM [48]	Align → Fuse	81.2	81.0	84.6	70.02	<u>79.2</u>	<u>78.5</u>
BERT-TL [61]	Align → Fuse	76.8	76.2	80.8	70.5	76.0	75.7
MER-HAN [50]	Align → Fuse	67.8	72.4	78.8	77.8	74.2	73.3
LFLB [51]	Align → Fuse	71.9	81.4	81.9	70.6	75.2	-
DBT [52]	Align → Fuse	63.9	81.9	82.7	67.4	74.0	72.2
MCFN [†] [53]	Align → Fuse	75.4	80.3	82.5	<u>71.4</u>	77.4	75.7
Ours (FEA-MER)	Fuse → Align	<u>79.5</u>	84.9	88.0	68.5	80.2	78.9
ΔSota	—	↓ 1.7	↑ 2.6	↑ 3.3	↓ 9.5	↑ 1.0	↑ 0.4

Notes: Bold values indicate the best performance. Underlined values denote the second-best performance. ΔSota represents the improvement of our method over the second-best baseline for each metric. (↑) indicates an improvement over the second-best performance, where higher values are better. (↓) indicates the absolute decrease relative to the best performance, where lower values are preferred. - indicates that the metric value was not provided in the original paper. [†] denotes results reproduced using the official implementation provided by the authors.

Table 2: Model Performance Comparison on the MELD

Methods	Align → Fuse Order	WACC	WF1
MCSCAN [54]	Align → Fuse	-	59.2
DIMMN [55]	Align → Fuse	60.6	58.6
SACMA [56]	Align → Fuse	62.3	59.3
MCFN [62]	Align → Fuse	64.5	62.2
SMCN [57]	Align → Fuse	64.9	62.3
SAMS [47]	Align → Fuse	65.4	62.6
RMERCT [58]	Align → Fuse	63.1	64.0
SMIN [59]	Align → Fuse	65.5	64.5
CARAT [†] [60]	Align → Fuse	<u>65.7</u>	<u>64.6</u>
Ours (FEA-MER)	Fuse → Align	66.5	65.1
ΔSota	—	↑ 0.8	↑ 0.5

Table 3: Ablation Study on the IEMOCAP and MELD Datasets

Settings	IEMOCAP		MELD	
	ACC	WACC	WACC	WF1
Text-only + MLP	71.4	69.9	63.1	61.8
Speech-only + MLP	74.5	73.0	54.2	51.0
Speech+Text + MLP	77.3	75.2	64.0	63.1
Cross-Attention Fusion	79.1	77.5	65.3	64.4
w/o STM	78.9	78.1	65.2	64.7
w/o EIA	79.3	77.8	64.8	64.2
w/o STM & EIA	78.4	76.1	64.5	63.8
Align → Fuse	79.4	78.0	65.7	64.3
w/o EIA Weighting	79.7	78.1	65.1	63.9
FEA-MER	80.2	78.9	66.5	65.1

limited improvements, highlighting the necessity of fusion mechanisms. Third, the fusion-first strategy outperforms the alignment-first variant, confirming our hypothesis that early alignment may

suppress modality-specific affective cues. Additionally, the "w/o EIA Weighting" experiment, where all positive and negative sample pairs are treated equally, results in a slight performance drop compared to the full model.

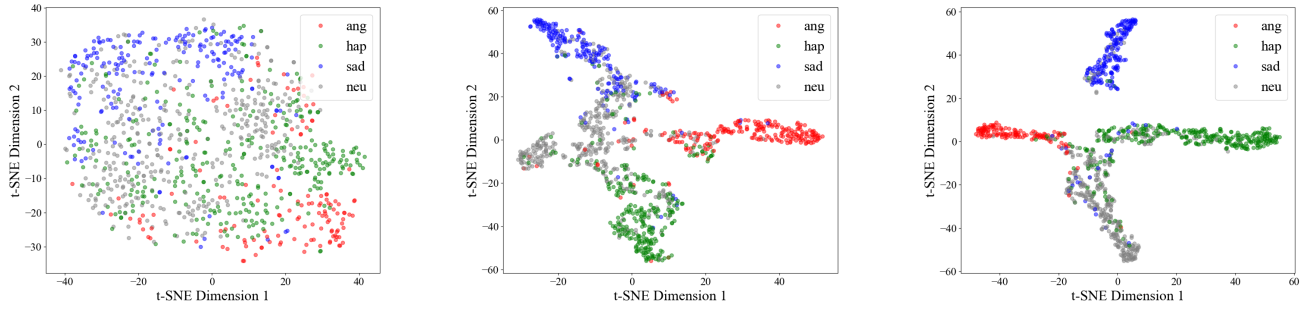
Notably, replacing self-attention fusion with bidirectional cross-attention yields competitive but inferior results. While cross-attention explicitly models inter-modal dependencies, it requires manually defined query directions, adding complexity. In contrast, our self-attention approach uses a learnable [CLS] token to capture global emotion representations, enabling symmetric intra- and inter-modal interaction without handcrafted queries, leading to better generalization and performance.

Moreover, the ablation results further demonstrate that the STM and EIA modules provide complementary modeling constraints for the fused representation by respectively preserving emotional differences between modalities and aligning cross-modal emotional semantics.

4.6 Research Analysis

4.6.1 (1) Is self-attention not as effective as cross-attention in multi-modal fusion? Cross-attention mechanisms are widely adopted in multimodal learning for their ability to explicitly model inter-modal interactions. However, they often depend on handcrafted query strategies and suffer from limited generalizability, especially under modality imbalance or noise. In contrast, our self-attention-based fusion strategy—applied over concatenated modality sequences with boundary-aware encoding and global aggregation tokens—enables uniform and symmetric modeling of both intra- and inter-modal dependencies, without the need for predefined query roles.

As shown in Table 3, replacing our self-attention fusion module with a bidirectional cross-attention structure yields competitive but inferior performance. This confirms that our proposed self-attention-based fusion strategy, as a whole, not only simplifies design and improves scalability, but also captures emotional interactions more effectively across modalities, outperforming more complex cross-attention-based alternatives.



(a) Fusion-only (no alignment): scattered and overlapping clusters.

(b) Align-then-fuse: better separation but partial mixing across modalities.

(c) Proposed FEA-MER: well-separated and compact emotion clusters.

Figure 3: T-SNE Visualization of Fused Emotion Representations from Different Fusion Strategies. Compared to direct fusion and alignment-first variants, FEA-MER achieves more discriminative and well-structured emotional clusters in the representation space.

4.6.2 (2) Why can Speech-Text Matching techniques improve the performance of emotion recognition? In multimodal settings, emotional signals from different modalities may be inconsistent or imbalanced, which can lead to biased fusion toward the dominant modality and result in suboptimal performance. STM explicitly regularizes the model to retain discriminative cues related to emotional consistency or inconsistency, thereby alleviating this issue. Specifically, the STM mechanism provides the following advantages:

(1) Enhancing the disentanglement of modality-specific emotional features: For emotionally inconsistent pseudo-paired samples, the STM guides the model to explicitly identify distinct emotional signals within each modality. This helps preserve modality-specific discriminative information, facilitates effective separation of shared and private features, and enhances the granularity and expressiveness of emotional representations.

(2) Facilitating modality collaboration and consistency modeling: For emotionally consistent pseudo pairs, the STM task uses label supervision to reinforce cross-modal collaborative perception and semantic fusion, thereby improving the model’s ability to align and express consistent emotional content. By jointly modeling both consistent and inconsistent samples, STM enables dynamic perception of cross-modal emotional relationships, thus improving the generalization and robustness of the primary emotion classification task.

4.6.3 (3) Why is Weighted Contrastive Learning based on continuous emotions effective? Although samples may belong to the same emotion category, they often differ significantly in terms of emotional intensity. However, existing contrastive learning approaches widely used in emotion recognition typically assume that all positive and negative sample pairs are equally important, neglecting emotional intensity as a crucial factor. This limitation hinders the model’s ability to capture fine-grained emotional semantics in the feature space. To address this issue, we propose a contrastive learning strategy weighted by emotional intensity. Specifically, we assign higher weights to positive pairs with small intra-class intensity differences and to negative pairs with large inter-class intensity differences.

This enables the model to jointly capture both the continuity of emotions and the boundaries between categories, thereby enhancing the discriminative power and alignment quality of emotion representations.

To further verify the superiority of our method in representation learning, we conduct t-SNE visualization of the final fused representations and compare them with those generated by the “no alignment” (fusion-only) and “align-then-fuse” variants. The visualization results are illustrated in Figure 3.

5 Conclusion

In this paper, we proposed FEA-MER, a novel multimodal emotion recognition framework that adopts a “fusion-first, alignment-later” strategy to overcome the limitations of conventional “alignment-first” approaches, which tend to suppress modality-specific emotional cues. FEA-MER consists of four key components: modality-specific encoding, a self-attention-based fusion module, emotion-aware optimization via Speech-Text Matching (STM) and Emotion Information Alignment (EIA), and a final classification head. Together, these modules enable the model to capture rich emotional structures and improve cross-modal synergy.

We conducted extensive experiments on IEMOCAP and MELD, demonstrating that FEA-MER consistently achieves competitive or state-of-the-art results. Ablation studies confirmed the individual contributions of each component, and visualization results further illustrated the effectiveness of our representation learning strategy.

It is worth noting that most existing multimodal emotion recognition datasets adopt majority-voting annotations without providing modality-specific emotion labels. This limitation hinders the model’s ability to understand the nuances of human emotional perception. In future work, we plan to explore and model the latent affective discrepancies across modalities, aiming to advance multimodal emotion recognition toward higher precision and more human-like understanding.

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